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10-11)

Goodness of Fit Test

We have N observations and we make K number of cells from them

C_1	C_2	C_3		C_N
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The goodness of fit test is based on following statistic:

$$\Rightarrow \chi^2 = \sum_{i=1}^K \frac{(O_i - E_i)^2}{E_i}$$

- O_i = observed freq of i^{th} cell
- E_i : expected freq of i^{th} cell
- K : number of cells
- $V = n - 1$: Degree of freedom

Rules:

- $\Rightarrow E_i$ should be at least 5, if not then we will combine it with adjacent cells.
- $\Rightarrow$ if O_i closely match E_i then χ^2 is small (indicating good fit)

Question 1: Fairness of a Die

A six-sided die is rolled **120 times**, and the observed outcomes are recorded in the following table: TEST GOODNESS OF FIT

Face	1	2	3	4	5	6
Observed	20	22	17	18	19	24

Example:

$$\alpha = 0.05 \quad N = 120 \quad P = \frac{1}{6}$$

Hypothesis:

H₀: The die is fair, distribution of outcomes follows uniform distribution.

H₁: The die is not fair, distribution of outcomes don't follow uniform distribution.

Test Stats: $e_i = \frac{1}{6} \times 120$

$$\chi^2 = \sum \frac{(O_i - e_i)^2}{e_i}$$

$$= \frac{(20-20)^2}{20} + \frac{(22-20)^2}{20} + \frac{(17-20)^2}{20}$$

$$+ \frac{(18-20)^2}{20} + \frac{(19-20)^2}{20} + \frac{(24-20)^2}{20}$$

$$= 1.7$$

Critical Region:

$$\chi^2_{0.05, \nu=5} = 11.070$$

(if $\chi^2 > \chi^2_{\alpha, \nu}$ reject H₀)

$$1.7 < 11.070$$

checking: $1.7 > 11.07$ (no)

H₀ Accept.

Conclusion:

Insufficient evidence to claim that die is not fair

Question 2: Testing Normality of Battery Lives A manufacturer claims that the **battery life** of a certain brand follows a **normal distribution** with a mean of **3.5 hours** and a standard deviation of **0.7 hours**. A random sample of **40 batteries** is tested, and the observed frequencies for different life ranges are recorded: TEST GOODNESS OF FIT

Battery Life Range (hours)	Observed (o_i)
1.45 - 1.95	2
1.95 - 2.45	1
2.45 - 2.95	4
2.95 - 3.45	15
3.45 - 3.95	10
3.95 - 4.45	5
4.45 - 4.95	3

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Data:
 $N = 40$
 $\mu = 3.5$
 $\sigma = 0.7$
 $\alpha = 0.05$

Solution:

Finding e_i

1) $e_1 = ?$ $P(X < 1.95) = P(Z < \frac{\bar{X} - \mu}{\sigma})$
 $= P\left(\frac{1.95 - 3.5}{0.7} < Z\right) = 0.0136$
 $e_1 = 40 \times 0.0136 = \boxed{0.5}$

2) $e_2 = ?$ $P(1.95 < X < 2.45)$
 $P\left(\frac{1.95 - 3.5}{0.7} < Z < \frac{2.45 - 3.5}{0.7}\right)$
 $P(Z < -2.21) - P(Z < -1.50)$
 $= 0.0668 - 0.0136$
 $= 0.0532$
 $e_2 = 40 \times 0.0532 = \boxed{2.1}$

3) $e_3 = ?$ $P(2.45 < X < 2.95)$
 $P\left(\frac{2.45 - 3.5}{0.7} < Z < \frac{2.95 - 3.5}{0.7}\right)$
 $= 0.148$
 $e_3 = 0.148 \times 40 = \boxed{5.9}$

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find e_4, e_5, e_6 with some method

$$7) e_7 = ? \quad P(X > 4.45)$$

$$P\left(\frac{X - 3.5}{7.0} > \frac{4.45 - 3.5}{7.0}\right) = 0.0677$$

$$e_7 = 0.0677 \times 40 = \boxed{3.5}$$

Table

Class Boundaries	O_i	e_i
1.45 - 1.95	2	0.5
1.95 - 2.45	1	2.1
2.45 - 2.95	4	5.9
2.95 - 3.45	15	10.3
3.45 - 3.95	10	10.7
3.95 - 4.45	5	7.0
4.45 - 4.95	3	3.5

We know that e_i should be at least 5. if not then we will combine it with adjacent e_i 's.

Applying formula

$$\chi^2 = \sum \frac{(O_i - e_i)^2}{e_i}$$

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$$\chi^2 = \frac{(7-8.5)^2}{8.5} + \frac{(15-10.3)^2}{10.3} + \frac{(10-10.7)^2}{10.7} \\ + \frac{(8-10.5)^2}{10.5}$$

$$\chi^2 = 3.05$$

Critical Region:

$n = 4$ (with respect to e.i's in table
after adjusting)

$$v = n - 1 = 3$$

$$\alpha = 0.05$$

$$\chi^2_{0.05(3)} = 7.815$$

$$\chi^2 > \chi^2_{0.05(3)}$$

checking

$$3.05 > 7.815 \text{ (No)}$$

No Accepted.

Question:

The following histogram was obtained by counting the occurrence of the first digits in telephone numbers in one column of a telephone directory:

digit	0	1	2	3	4	5	6	7	8	9
observed	0	0	24	2	25	3	32	15	2	2

Test the goodness of fit of this data to a random variable that is uniformly distributed in the set $\{0, 1, \dots, 9\}$ at a 1% significance level.

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Question 5:

Data:

Digit :	0	1	2	3	4	5	6	7	8	9
observed:	0	0	24	2	25	3	32	15	2	2

$\alpha = 1\%$

To find:

Test the goodness of fit of data to RV that is uniformly distributed.

Solution:

$\Rightarrow$ Hypothesis:

H_0 : The first digits are uniformly distributed.

H_1 : The first digits are not uniformly distributed.

$\Rightarrow$ Test Statistics:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad \text{--- ①}$$

Total number of obs

$$0 + 0 + 24 + 2 + 25 + 3 + 32 + 15 + 2 + 2 = 105.$$

as they are uniformly distributed

$$E_i = \frac{N}{n} = \frac{105}{10} = 10.5$$

Put all values in ①

$$\chi^2 = \frac{(0-10.5)^2}{10.5} + \frac{(0-10.5)^2}{10.5} + \frac{(24-10.5)^2}{10.5} + \frac{(2-10.5)^2}{10.5}$$

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$$\frac{(25-10.5)^2}{10.5} + \frac{(3-10.5)^2}{10.5} + \frac{(32-10.5)^2}{10.5} + \frac{(15-10.5)^2}{10.5} \\ + \frac{(2-10.5)^2}{10.5} + \frac{(2-10.5)^2}{10.5}$$

$$\chi^2 = 10.5 + 10.5 + 17.36 + 6.88 + 20.02 + 5.36 \\ + 44.02 + 1.93 + 6.88 + 6.88 + 130.33$$

$$\boxed{\chi^2 = 130.33}$$

⇒ Critical Region

$$df = v = n-1 = 10-1 \\ v = 9$$

$$\alpha = 0.001$$

value at $v=9$ with $\alpha=0.001$ is
21.66

$$21.66 < 130.33$$

Reject H_0

⇒ Conclusion: The statistical value is inside the critical region, so we cannot say that the first digits are uniformly distributed.